测试报告 报告编号：A11240410023 日期:2024年12月26日

委托单位： 额敏县金宇泉塑料制品有限公司
新疆塔城地区额敏县塔额路十六地段 020100075-2 号

产品名称： 全生物降解地膜

型号规格： 700*0.008;农用地膜薄膜Ⅲ类

检测类别： 委托检测

样品名称：	全生物降解地膜		
商标：	金宇泉	规格型号等级：	700*0.008 农用地膜薄膜Ⅲ类
生产日期：	2024年3月1日	样品数量：	600G
生产单位：	额敏县金宇泉塑料制品有限公司		
样品接收日期：	2024年04月11日	报告发放日期：	2024年12月26日
测试周期：	2024年04月17日至2024年08月14日	样品参考信息：	/
检测依据：	GB/T 19277.1-2011 受控堆肥条件下材料最终需氧生物分解能力的测定 采用测定释放的二氧化碳的方法 第1部分：通用方法 GB/T 1040.3-2006 塑料 拉伸性能的测定 第3部分:薄膜和薄片的试验条件 GB/T 1130-1991 塑料 直角撕裂性能试验方法 GB/T 1037-1988 塑料薄膜和片材透水蒸气性试验方法 杯式法 GB/T 15337-2008 原子吸收光谱分析法通则		
判定依据：	GB/T 35795-2017 全生物降解农用地面覆盖薄膜 5.6生物降解性能		
检测结论：	符合。详见本报告检测结果		
备注：	/		

编制： 洪欣欣 审核： 黄嘉欣 批准：

安瑞检测认证(广东)有限公司 检验检测专用章

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序号	检 测 项 目	技术指标	检测结果	单向结论
01	厚度极限偏差，mm	±0.003	-0.002～+0.001	合格
02	厚度平均偏差，%	+15 -12	-7 -8	合格
03	外观	不允许有影响使用的气泡、斑点、折褶、杂质和针孔等缺陷，对不影响使用的缺陷不得超过20个/100cm²	无影响使用的气泡、斑点、折褶、杂质和针孔等缺陷，100cm²种不影响使用的缺陷为5个	合格
04	拉伸负荷 纵/横向，N	≧2.00/≧2.00	3.50/2.73	合格
05	断裂标称应变 纵/横向，%	≧150/≧250	231/733	合格
06	直角撕裂负荷 纵/横向，N	≧0.80/≧0.80	1.00/1.49	合格
07	水蒸气通过量，g/(m²·24h)	<800	763	合格
08	重金属含量			
	砷(As),mg/Kg	≦5	0.266	合格
	镉(Cd),mg/Kg	≦0.5	0.228	合格
	铜(Cu),mg/Kg	≦50	<0.02	合格
	铅(Pb),mg/Kg	≦50	1.58	合格
	汞(Hg),mg/Kg	≦0.5	<0.0003	合格
	镍(Ni),mg/Kg	≦25	3.75	合格
	硒(Se),mg/Kg	≦0.75	<0.0003	合格
	锌(Zn),mg/Kg	≦150	<0.02	合格
	铬(Cr),mg/Kg	≦50	0.877	合格
	钼(Mo),mg/Kg	≦1	0.399	合格
	氟(F),mg/Kg	≦100	15.4	合格
	钴(Co),mg/Kg	≦38	<0.0005	合格
09	人工气候老化性能 老化100h后断裂标称应变 纵/横向，%	≧80/≧100	262/494	合格
10	生物降解性能 有机成分,%	≧51	92.2	合格
11	相对生物分解率(120d),%	≧90	93.7	合格

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1 摘要

本试验检测了样品在受控堆肥条件下的最终需氧生物分解能力。在试验体系中，以样品作为的有机碳源，用无二氧化碳的空气以受控速率对试验混合物进行曝气。通过测定二氧化碳产生量来确定降解率。

在本试验条件下：

试验 45 天后，参比材料微晶纤维素的平均生物分解百分率是 73.92%；

测试样品（样品名称：全生物降解地膜）120 天的平均生物分解百分率为 90.11%，相对生物分解率为 93.77%。

依据 GB/T 35795-2017 相对生物分解率要求≥90%，该产品符合要求。

本试验过程满足如下条件：

试验前 10 天空白容器内产生的二氧化碳量的平均值为 119.80mg CO_2/g 挥发性固体，在 50-150mg 范围内；

试验第 45 天参比材料微晶纤维素平均生物分解百分率为 73.92%，大于 70%。

根据 GB/T 19277.1-2011 对结果有效性规定，本试验有效。

2 引言

2.1 试验名称受控堆肥条件下材料最终需氧生物分解能力的测定。

2.2 试验目的测定样品在受控堆肥条件下的最终需氧生物分解和崩解程度。

2.3 原理本测定方法在模拟的强烈需氧堆肥条件下，测定试验材料最终需氧生物分解能力和崩解程度。使用的培养土来自于稳定的、腐熟的堆肥，如可能，从城市固体废弃物中有机物的堆肥中获取。试验材料与培养土混合，导入静态堆肥容器。在该容器中，混合物在规定的温度、氧浓度和湿度下进行强烈的需氧堆肥。试验周期不超过 6 个月。在试验材料的需氧生物分解过程中，二氧化碳、水、矿化无机盐及新的生物质都是最终生物分解的产物。在试验中连续监测、定期测量产生的二氧化碳，累计产生的二氧化碳量。试验材料在试验中实际产生的二氧化碳量与该材料可以产生的二氧化碳的理论量之比为生物分解百分率。根据实际测量的总有机碳(TOC)含量可以计算出二氧化碳的理论释放量。生物分解百分率不包括

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已转化为新的细胞生物质的碳量，因为它在试验周期内不代谢为二氧化碳。

2.4 试验起止日期

接样日期：2024 年 04 月 11 日

检测启动日期：2024 年 04 月 17 日

检测完成日期：2024 年 08 月 14 日

2.5 检测机构信息

名称：安瑞检测认证（广东）有限公司

地址：东莞市厚街镇湖景大道高新科技园

电话：86-769-8258 9333

2.6 参加试验人员

检测负责人：陈海

检测人员：李均华、陈旭文

3 试验材料

3.1 样品名称：全生物降解地膜

性状：透明塑胶

保存条件： /

上述信息由委托方提供。

3.2 试验条件

试验于 58℃±2℃恒温，弱光，没有任何会影响微生物生长的蒸汽条件下进行。

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3.3 仪器设备

(1)反应装置；

(2)压缩空气制备装置；

(3)气体吸收装置；

(4)气体流量控制器；

(5)恒温装置。

4 试验前准备工作

4.1 接种物准备

自制堆肥接种物，除去杂质后通过 0.5cm 的筛子。

经测定，本试验所用堆肥：

水分含量 46.26%；总干固体含量 53.74%；挥发性固体含量 19.31%；

pH 值 7.65；总碳含量/%；总氮含量/%。

4.2 吸收瓶使用

每个吸收瓶中加入 400mL20g/L 的氢氧化钠溶液作为吸收剂，每个反应容器配备串联的三个吸收瓶。

4.3 反应容器的数量

3 个装试验材料的容器；

3 个装阳性参比材料的容器；

3 个空白容器。

4.4 样品的准备

对样品进行粉碎处理。经测定，各项指标如下：

样品水分含量 0.07%；总碳含量 53.79%。

按照接种物与试验样品的干重比约为 6:1 的比例进行混合制备样品试验体系，加 300g 接种物（干重）和 50g 样品（干重）到反应容器中进行试验。

4.5 参比物的准备

对参比物进行测定。参比材料水分含量为 0.04%，总碳含量为 42.36%。

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5 试验方法

（1）用无 CO_2 的湿度饱和空气以恒定流量对试验体系进行曝气，于 58℃±2℃条件下进行试验。

（2）定期测量堆肥容器排出气中的氧气浓度，第一星期每天测定 2 次，以后，测量次数可以减少，必要时，调节空气流量。

（3）CO_2 的测定：

用氢氧化钠吸收法：拆出直接连接防倒吸装置的氢氧化钠吸收瓶来测定，其余两个吸收瓶顺序前移连接至防倒吸装置，并在末端连接上一个新配制的氢氧化钠吸收瓶。拆下的吸收瓶用 HCl 标准溶液进行滴定。在开始的 45 天里，每天测定一次 CO_2 的吸收量，45 天后每周测量 2 次 CO_2 的吸收量。

（4）每天对试验装置进行气密性检查。

（5）每周对试验容器进行振荡混匀，同时测定混合物的 pH 值，并观察混合物的性状：如试验容器排放气的气味，堆肥湿度、色泽、霉菌生成，试验样品结构变化、崩解程度等。

6 结果有效性判断

6.1 45 天后参比材料的生物分解率超过 70%（实际结果：73.92%）

6.2 前 10 天内，空白容器中堆肥接种物产生 50-150mg CO2/g 挥发性固体（实际结果：119.80mg）

6.3 试验结束时每个堆肥容器的生物分解百分率之间的相对偏差不超过 20%。

（1)参比材料 3 个反应瓶在试验结束时(第 120 天)的生物分解百分率分别为:93.58%,98.44%, 96.28%；3 个反应瓶的生物分解百分率相对偏差＜20%。

（2)试验样品 3 个反应瓶在试验结束时(第 120 天)的生物分解百分率分别为:90.75%,90.08%, 89.50%；3 个反应瓶的生物分解百分率相对偏差＜20%。

7 数据结果

7.1 生物分解率结果摘要：

样品类别	10 天二氧化碳释放量/挥发性固体(mg/g)	45 天生物分解率（%）	试验结束时（120 天）最终生物分解率（%）	试验结束时（120 天）相对生物分解率（%）
样品	/	/	90.11	93.77
参比材料	/	73.92	96.10	/
空白	119.80	/	/	/

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7.2 测试过程中二氧化碳累计产生量及降解率：

时间（天）	平均累计 CO_2 产生量（g/容器）			平均累计降解率（%）	
	空白容器	参比材料	试验材料	参比材料	试验材料
1	0.77	0.86	0.87	0.13	0.11
2	1.46	1.80	1.92	0.43	0.46
3	2.23	2.71	2.87	0.62	0.65
4	2.93	3.68	3.97	0.97	1.06
5	3.53	4.59	5.25	1.36	1.74
6	4.25	5.56	6.81	1.69	2.60
7	4.86	6.60	7.81	2.24	3.00
8	5.50	7.60	9.34	2.70	3.89
9	6.23	8.77	10.41	3.27	4.24
10	6.94	9.93	11.50	3.86	4.63
12	8.33	12.26	13.65	5.06	5.40
14	9.61	14.77	15.75	6.64	6.23
16	10.88	17.55	17.88	8.60	7.11
18	12.24	20.82	20.00	11.05	7.87
20	13.63	24.45	22.19	13.94	8.69
22	15.10	28.42	24.53	17.15	9.57
24	16.44	32.67	26.64	20.90	10.34
26	17.92	37.27	28.80	24.92	11.03
28	19.27	42.53	31.00	29.95	11.90
30	20.68	48.01	33.23	35.19	12.72
32	22.03	53.86	35.44	40.98	13.59
34	23.40	60.02	37.78	47.14	14.58
36	24.88	66.14	40.00	53.14	15.34
38	26.39	71.93	42.39	58.64	16.23
40	28.00	77.26	44.52	63.44	16.76
42	29.61	82.32	46.72	67.87	17.34
44	31.13	87.06	48.99	72.03	18.12
45	32.11	89.52	50.10	73.92	18.24
47	33.37	93.33	52.17	77.21	19.06
49	34.70	96.69	54.55	79.82	20.13
51	36.16	99.62	57.24	81.72	21.38

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	空白容器	参比材料	试验材料	参比材料	试验材料
53	37.71	102.21	60.12	83.06	22.73
55	39.33	104.61	63.46	84.06	24.46
57	40.94	106.83	67.05	84.84	26.48
59	42.55	108.89	70.74	85.43	28.59
61	44.13	110.89	75.04	85.96	31.35
63	45.83	112.91	79.53	86.38	34.18
65	47.44	114.88	84.50	86.84	37.58
67	49.18	116.81	89.78	87.08	41.17
69	51.00	118.69	95.09	87.15	44.70
71	52.97	120.65	100.22	87.15	47.91
73	55.28	122.61	105.48	86.69	50.90
75	57.37	124.52	110.52	86.47	53.90
77	59.41	126.50	115.37	86.39	56.75
79	61.42	128.47	119.69	86.33	59.08
81	63.29	130.41	123.91	86.43	61.47
83	65.12	132.37	127.71	86.59	63.47
85	67.03	134.36	131.18	86.71	65.05
87	68.88	136.36	134.40	86.89	66.45
89	70.61	138.30	137.31	87.17	67.65
90	71.56	139.60	138.72	87.61	68.11
92	73.04	141.53	141.64	88.19	69.56
94	74.46	143.40	144.46	88.78	70.98
96	75.81	145.16	147.26	89.30	72.45
98	77.05	146.85	150.00	89.87	73.97
100	78.33	148.47	152.77	90.31	75.49
102	79.54	150.09	155.52	90.85	77.05
104	80.73	151.79	158.19	91.50	78.54
106	81.86	153.42	160.80	92.14	80.05
108	82.92	154.96	163.50	92.77	81.71
110	84.02	156.34	166.12	93.12	83.25
112	85.02	157.80	168.61	93.72	84.76
114	86.01	159.23	171.07	94.28	86.25

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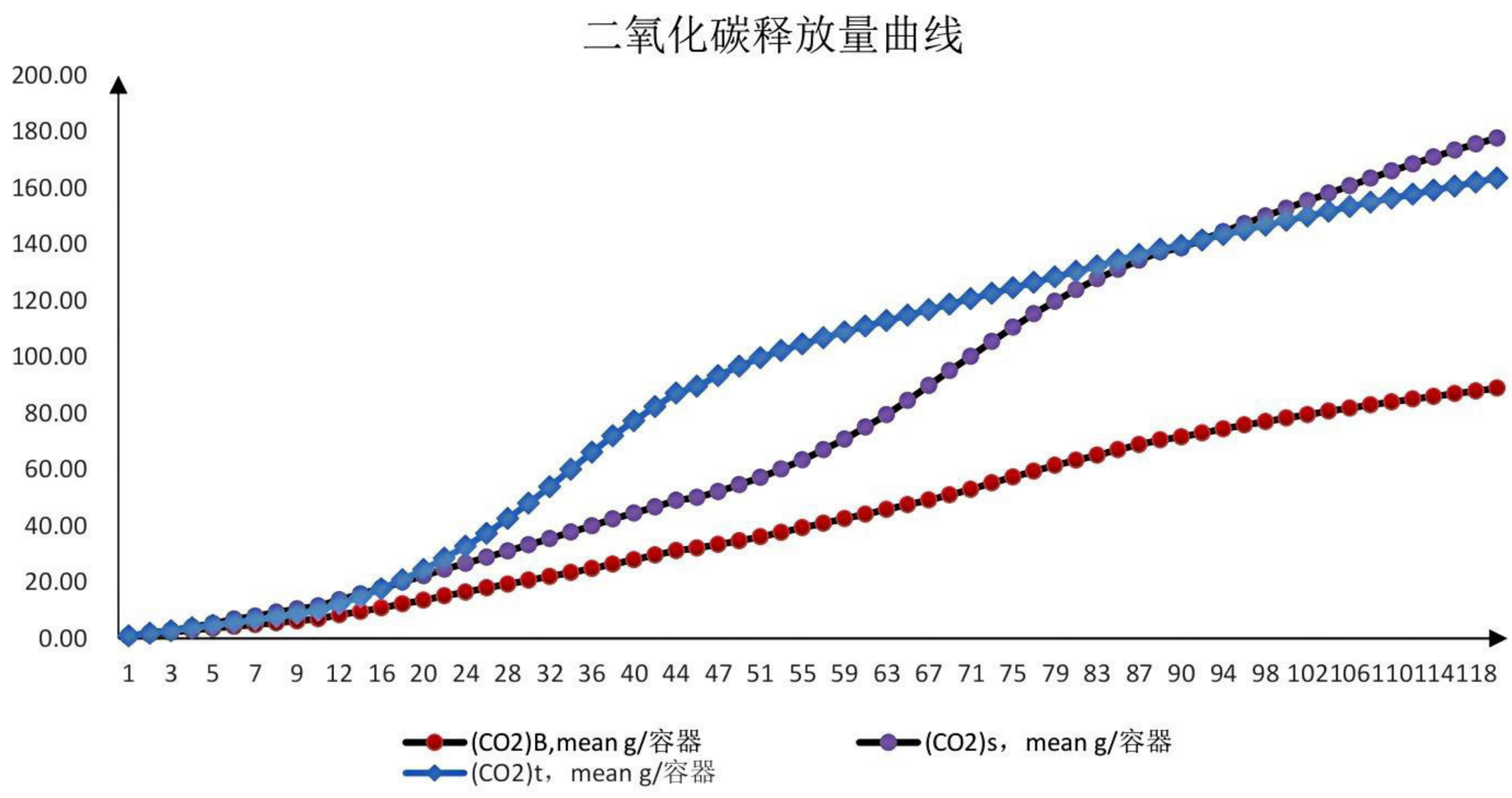
时间（天）	平均累计 CO_2 产生量（g/容器）			平均累计降解率（%）	
	空白容器	参比材料	试验材料	参比材料	试验材料
116	86.99	160.66	173.45	94.86	87.67
118	87.92	162.13	175.67	95.56	88.99
120	88.92	163.55	177.78	96.10	90.11

7.3 时间-二氧化碳释放量（平均值）曲线图



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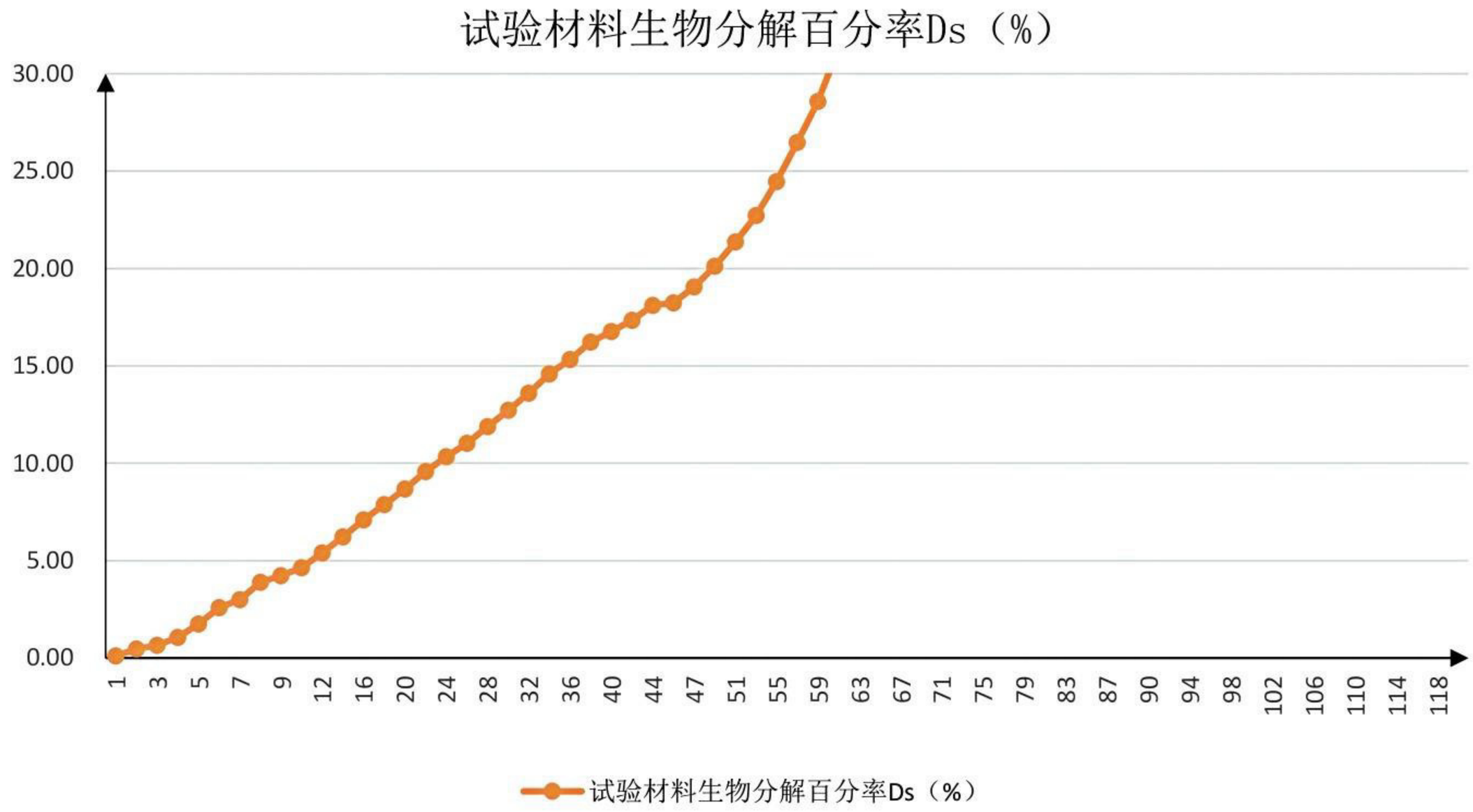
7.4 试样材料的时间-生物分解百分率（平均值）曲线图



8 依据

国家标准 GB/T 19277.1-2011，受控堆肥条件下材料最终需氧生物分解能力的测定采用测定释放的二氧化碳的方法第 1 部分：通用方法。

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样品照片:

试验前 样品照片

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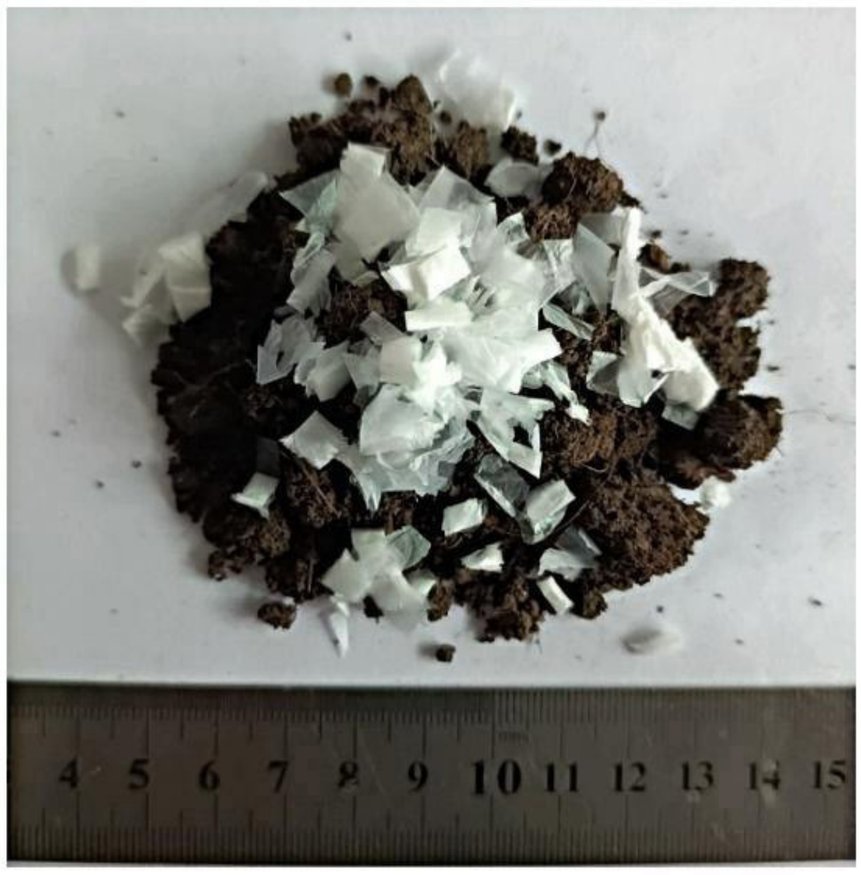

试验前 样品与堆肥接种物照片

试验后（第120天） 样品与堆肥接种物照片

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